FitBuddy – AI Fitness Plan Generator using Gemini Models

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Activity 1.2 – Define the Architecture of the System

Introduction:

The system architecture defines the overall structure of the FitBuddy application. It explains how different components of the system interact with each other to generate personalized fitness plans using Artificial Intelligence.

FitBuddy uses AI models to analyze user input such as age, fitness goals, weight, and lifestyle to generate a customized workout and fitness plan. A well-defined system architecture ensures that the application works efficiently, securely, and smoothly.

Objective of the Systems Architecture

The main objective of defining the system architecture is to organize the system into structured components so that developers can easily build and maintain the application.

The architecture helps to:

- Manage system components efficiently
- Improve performance and reliability
- Enable AI-powered fitness recommendations
- Ensure proper data handling
- Make the application scalable for future users

Components of FitBuddy System Architecture

The FitBuddy system consists of several important components that work together to provide personalized fitness plans.

1. User Interface (UI)

The User Interface is the front-end part of the application where users interact with the system.

Functions:

- Collect user information
- Display workout plans

- Show fitness recommendations
- Allow users to update their fitness goals

Examples:

- Login page
- User profile page
- Fitness goal selection page
- Workout plan dashboard

The UI must be simple, responsive, and easy to use so that users can easily interact with the application.

2. Application Logic Layer

This layer contains the main processing logic of the FitBuddy system.

Functions:

- Process user input
- Communicate with AI models
- Generate personalized fitness plans
- Handle application operations

For example, when a user enters their fitness goal (such as weight loss or muscle gain), the system processes this information and sends it to the AI model to generate a suitable workout plan.

3. AI Processing Layer (Gemini Model Integration)

This is one of the most important components of FitBuddy.

The system integrates AI models such as Google Gemini to generate intelligent fitness plans.

Functions:

- Analyze user fitness data
- Generate workout suggestions
- Recommend diet and exercise routines
- Adapt plans based on user progress

Using AI allows the application to provide personalized and intelligent fitness guidance.

4. Database Layer

The database stores all important information related to the users and their fitness plans.

- Stored information includes:
- User profiles
- Fitness goals

- Generated workout plans
- User progress data
- The database ensures that data is stored securely and can be retrieved whenever required.

System Workflow of FitBuddy

The FitBuddy system follows a simple workflow:

Step 1

The user opens the application and enters personal details such as age, weight, fitness goals, and experience level.

Step 2

The application sends this information to the system processing layer.

Step 3

The system communicates with the AI model to generate a personalized fitness plan.

Step 4

The AI analyzes the data and creates a customized workout routine.

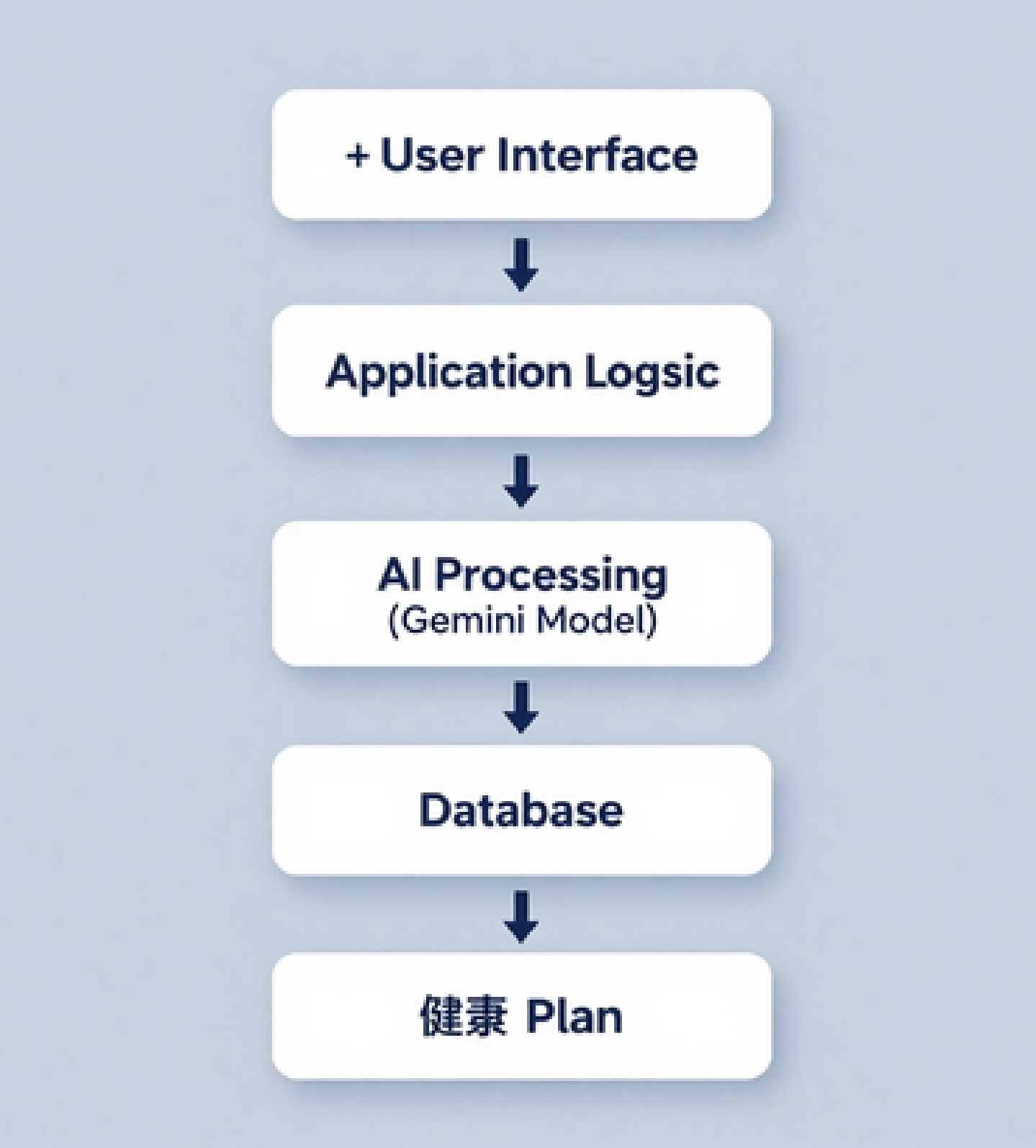
Step 5

The generated fitness plan is sent back to the application.

Step 6

The user can view their workout plan on the dashboard.

Architecture Diagram



Advantages of the FitBuddy Architecture

>>> The architecture provides many benefits:

- Personalized Fitness Plans:

The AI model generates customized workout plans for each user.

- Efficient System Performance:

The layered architecture improves system efficiency.

- Scalability:

The system can support many users in the future.

- Data Security:

User information is stored securely in the database.

- Easy Maintenance:

Developers can easily update and improve different parts of the system.